Sanjana Shivanand

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EDUCATION

Binghamton University, SUNY – Binghamton, NY *Expected December 2026*
Master of Science in Computer Science | GPA: 3.50/4.00

JSS Academy of Technical Education, Bangalore, India
Bachelor of Engineering in Computer Science *June 2024*

SKILLS

Programming & Data Engineering: Python, SQL, PySpark, ETL, Data Pipelines, Batch and Incremental Processing, Data Integration, Data Modeling, Schema Mapping, Data Validation, Data Quality Testing
Analytics & BI: Power BI, DAX, Tableau, Excel, Power Query, KPI Development, Dashboard Development, Exploratory Data Analysis, Time-Series Analysis
Machine Learning, NLP & GenAI: scikit-learn, XGBoost, PyTorch, Hugging Face, FinBERT, MLflow, Regression, Classification, Forecasting, Anomaly Detection, LangChain, LangGraph, RAG, Qdrant, LLM Agents
Cloud, Databases & Tools: AWS S3, Athena, Glue, Lambda, PostgreSQL, MySQL, SQLite, MongoDB, Apache Airflow, Prefect, dbt, Docker, Git, FastAPI, Django, REST APIs, Power Automate

WORK EXPERIENCE

Metropolitan Transportation Authority | Staten Island, USA *January 2026 - Present*
Data Analyst Intern

- Created data-pipeline workflows using SQL, Python and Power Query to clean and integrate Policy Compliance, Employee training and safety data into standardized datasets for Power BI dashboards.
- Streamlined employee disciplinary-action and injury reporting, reducing report-generation time by approximately 60%, by automating data collection, validation and report identifiers with Power Automate.
- Implemented rule-based validation and scheduling logic for 600+ annual training assignments, enforcing employee eligibility, session capacity, timing conflicts and duplicate prevention.

Centre for Development of Telematics | Bangalore, India *January 2024 – October 2024*
Data Analyst & Machine Learning Intern

- Improved data readiness for O-RAN 5G Network analysis by building Python and Django workflows to upload, merge, clean and transform datasets through null handling, duplicate checks, scaling and encoding.
- Enabled model comparison for network slicing and V2X handover use cases by integrating classification, regression and LSTM workflows and evaluating results using accuracy, precision, recall, F1 score and MSE,
- Made model insights easier to interpret by developing Django reporting and visualization features for dataset metadata, model-performance comparisons and prediction results.

PROJECTS

SafeEats NYC - Restaurant Safety Analytics Platform | *GitHub* *July 2026 - Present*

- Built an automated ETL pipeline for nearly 400K NYC inspection and 311 records using Python, PySpark, Airflow, Amazon S3, and geospatial matching.
- Designed an Athena and dbt dimensional warehouse with reusable fact tables, dimensions, KPI marts, data-quality tests, and cross-layer reconciliation.
- Developed a leakage-safe XGBoost model that scored 24K restaurants for future B/C-grade risk through temporal validation, feature engineering, and probability calibration.

FinSight AI - Financial Intelligence Platform | *GitHub* *February 2026 - May 2026*

- Built a Prefect-orchestrated daily pipeline that ingests price, news, SEC filing and macroeconomic data for 25 assets across five classes from Yahoo Finance, NewsAPI, and FRED transforming it through dbt and DuckDB models with automated data-quality tests.
- Evaluated MLflow-tracked HAR-RV volatility forecasts where it outperformed its naive baseline for 22 assets, Prophet price forecasts, FinBERT sentiment scoring and Isolation Forest anomaly detection.

Social Media Intelligence Platform | *GitHub* *September 2025 – December 2025*

- Built continuous ingestion pipelines for five 4chan boards and six Reddit communities, processing 3.6M+ 4chan posts and 50K+ Reddit records into PostgreSQL and TimescaleDB.
- Analyzed cross-platform activity trends, sentiment and toxicity using SQL time-series analysis, VADER sentiment scoring and Google Perspective API.
- Developed FastAPI and Next.js analytics interfaces with a LangChain SQL agent, enabling users to explore behavioral trends and query stored social-media data in natural language.